Entra ID configuration guide

Important notice

Before proceeding with this configuration guide, you must first complete the Microsoft Entra ID application creation process. This includes creating an app registration, configuring API permissions, and generating client secrets.

Please refer to the Microsoft Entra ID App creation for Radiator Policy Server authentication before continuing.

Example configuration setup

The Radiator installation includes an Entra ID example configuration that demonstrates authentication with Microsoft Entra ID and MFA support, which is a good starting point. To copy the example configuration as the working configuration, run these commands in your terminal:

```
# Navigate to the Radiator configuration directory
cd /var/lib/radiator/

# Copy all Entra ID example files to your working configuration
cp -R /opt/radiator/server/doc/example-configurations/entra-auth-mfa/* .
```

Please note that the copied configuration contains placeholder values that must be replaced with your actual Entra ID application information (covered in the following sections).

Service principal creation

Microsoft Entra ID requires a service principal with appropriate permissions to handle MFA notifications. With the powershell script below, you can create a new client secret for Microsoft's Entra ID MFA Notification Client. This secret has to be updated to the Radiator configuration file policies.conf so that Radiator is allowed to use the MFA notification service (covered in the next section).

Run this PowerShell script to create the required secret (running this script requires global admin rights):

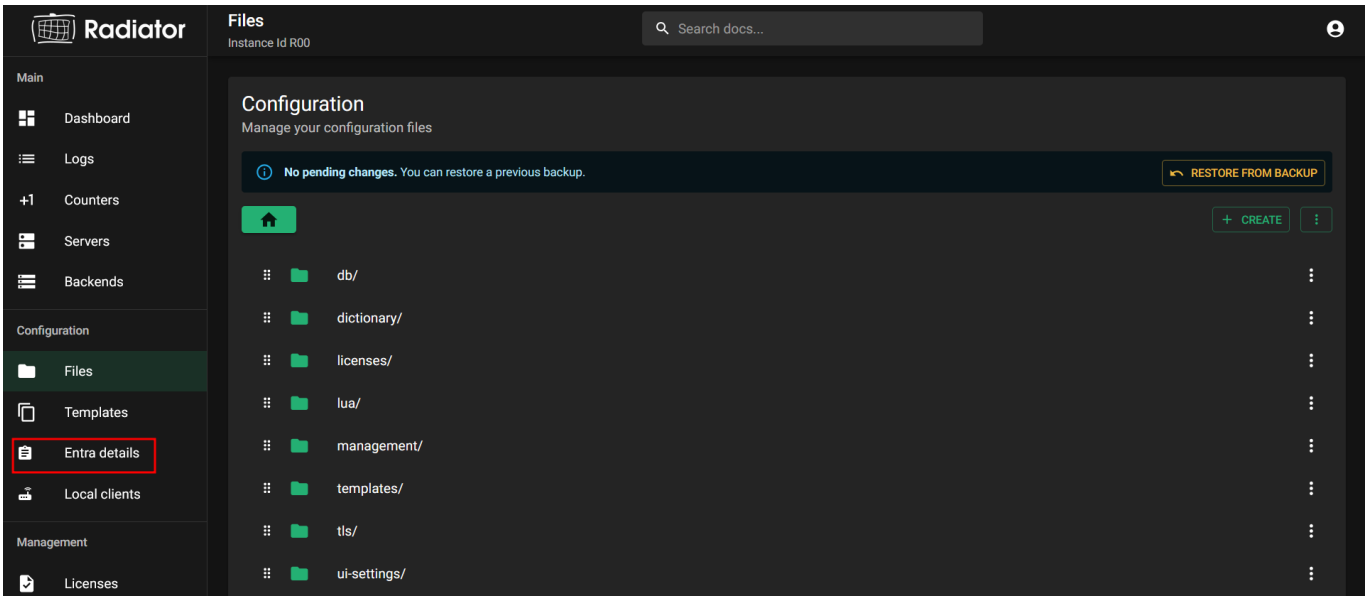
```
Connect-MgGraph -Scopes 'Application.ReadWrite.All'
# Get ID of 'Entra Id MFA Notification Client' Service Principal
$servicePrincipalId = (Get-MgServicePrincipal -Filter "appid eq '981f26a1-
7f43-403b-a875-f8b09b8cd720'").Id

$params = @{
    passwordCredential = @{
        displayName = "My Application MFA"
    }
}

# Create Client Secret onto client
```

```
$secret = Add-MgServicePrincipalPassword -ServicePrincipalId
$servicePrincipalId -BodyParameter $params
Write-Output $secret
```

Record the secret's value for use in Radiator Policy Server configuration.

Updating the example configuration

After completing previous steps as well as the 'Microsoft Entra ID App creation for Radiator Policy Server authentication' guide, you will need to modify the example configuration from the first step to match your specific Entra ID application details. This can be done from the GUI by selecting **Entra details** on the left side panel and filling in the fields (see below).



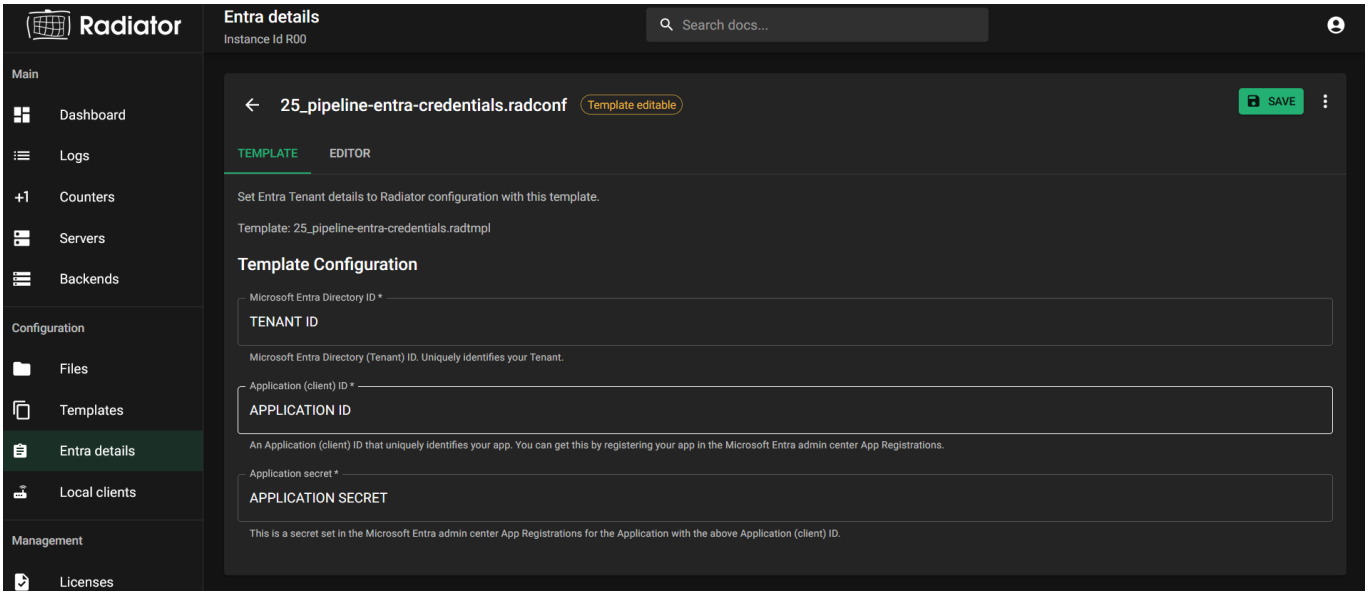
Alternatively use command line (CLI). Start by opening `/var/lib/radiator/25_pipeline-entra-credentials.radconf`:

```
sudo vi /var/lib/radiator/25_pipeline-entra-credentials.radconf
```

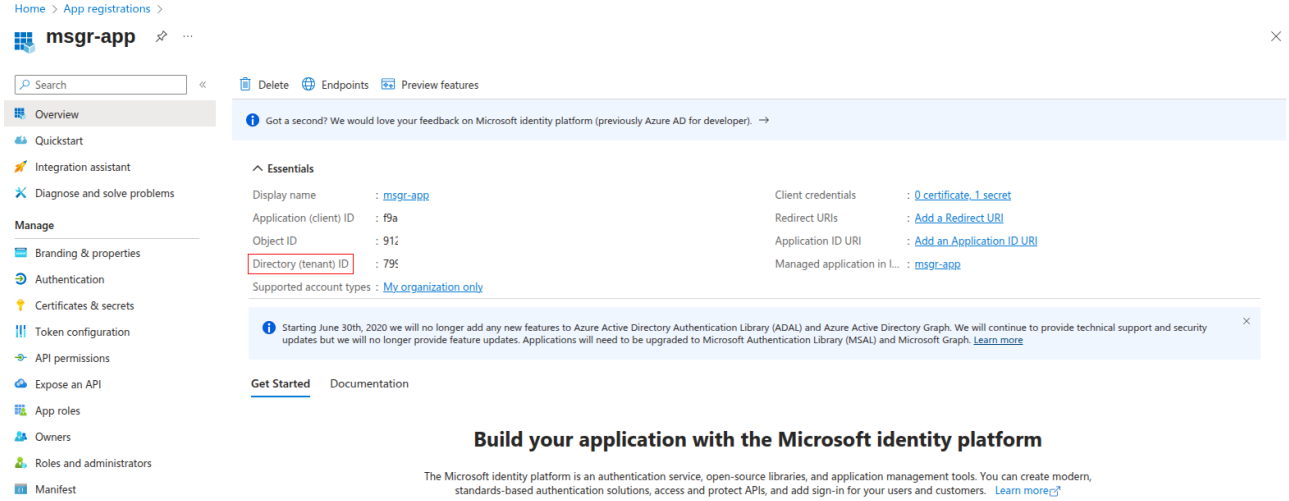
In `25_pipeline-entra-credentials.radconf`, locate the pipeline block named "SET_TENANT_INFORMATION" which looks something like this:

```
aaa {
    pipeline "SET_TENANT_INFORMATION" {
        modify vars.aaa.oauth2.token-url =
"https://login.microsoftonline.com/TENANT ID/oauth2/v2.0/token";
        modify vars.aaa.oauth2.client = "APPLICATION ID";
        modify vars.aaa.oauth2.secret = "APPLICATION SECRET";
        modify vars.mfa.oauth2.token-url =
"https://login.microsoftonline.com/TENANT ID/oauth2/token";
        modify vars.mfa.oauth2.secret = "MFA PROVISIONING SECRET";
    }
}
```

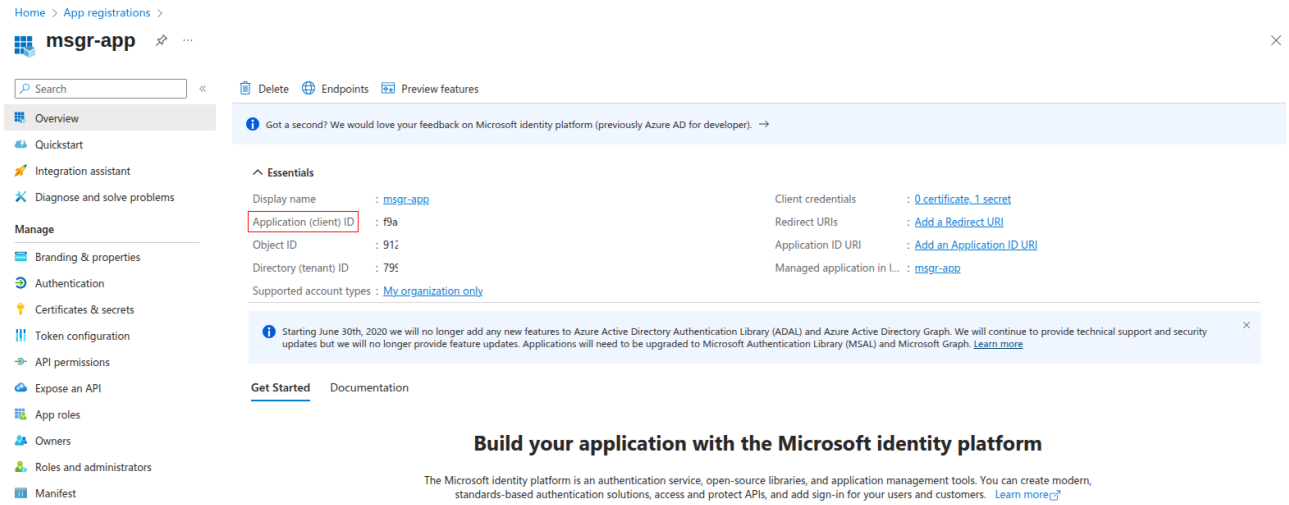
Update the following values either in the GUI or in CLI:



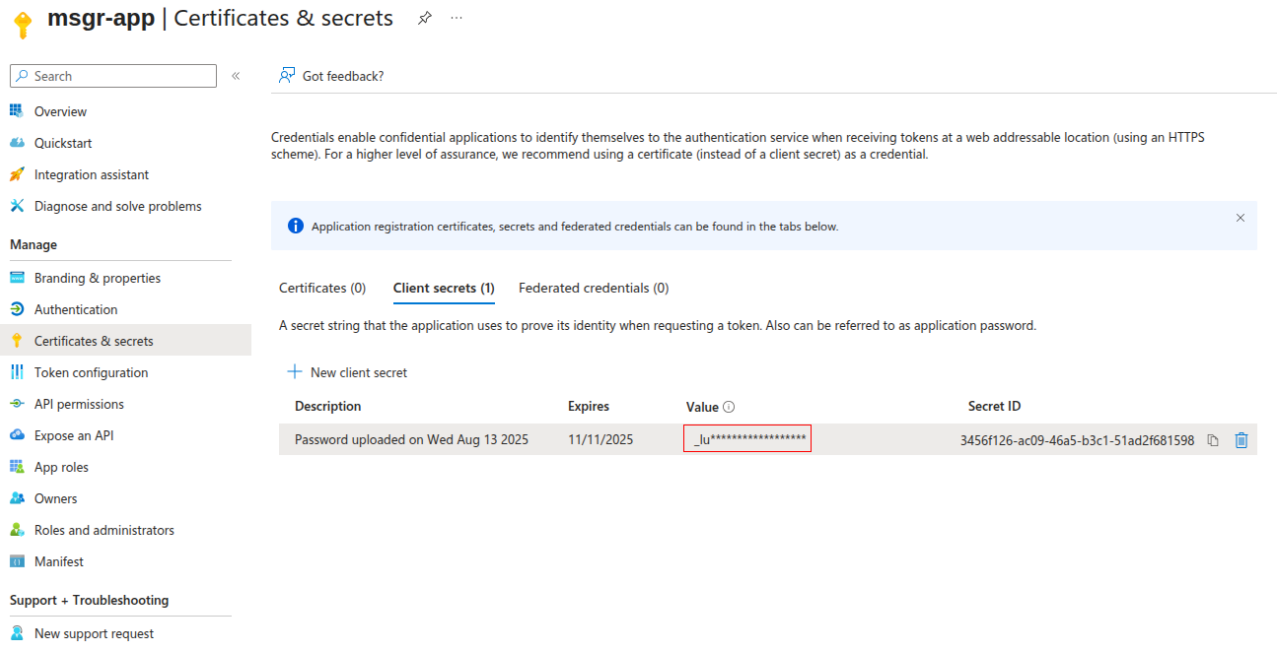
1. In GUI replace `TENANT ID` and in CLI in vars.mfa.oauth2.token-url and vars.aaa.oauth2.token-url, replace `<Directory (tenant) ID>` with the Directory (tenant) ID. The Directory (tenant) ID is shown on the application Overview page:



2. In GUI replace `APPLICATION ID` and in CLI in vars.aaa.oauth2.client, replace `<Application (client) ID>` with the Application (client) ID. The Application (client) ID is shown on the application Overview page:



3. In GUI replace `APPLICATION SECRET` and in CLI in vars.aaa.oauth2.secret, replace `<Application secret>` with the secret. The secret is shown on the application Certificates & secrets page right after creation:



4. In GUI replace `MFA PROVISIONING SECRET` and in CLI in vars.mfa.oauth2.secret, replace `<Secret from provisioning script>` with the secret you created in the service principal creation section.
5. Save the configuration file by pressing **Create** on the GUI or using your editor's Save on the CLI.

When doing changes on the GUI, new configuration is automatically taken into use. On the CLI, restart Radiator Policy Server:

```
sudo systemctl restart radiator-server
```

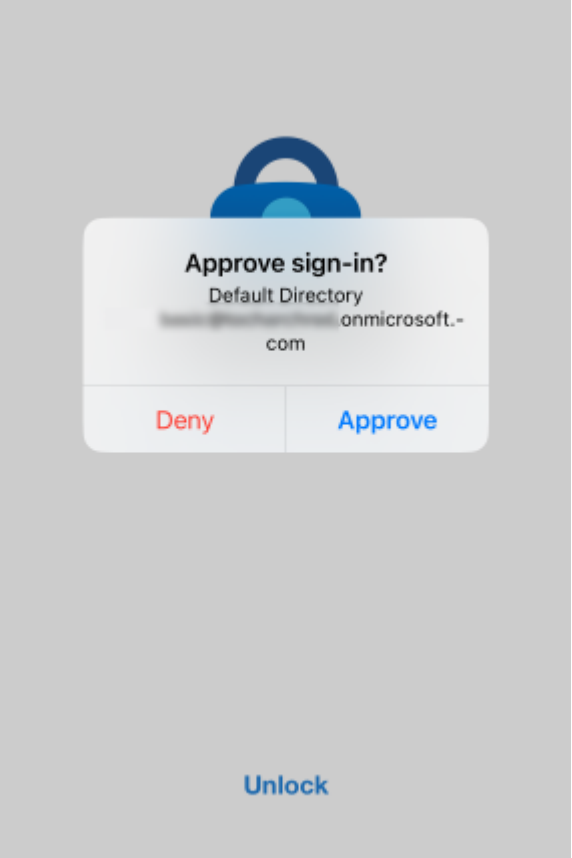
Testing

You can test the Entra ID authentication using the Radiator test client on the CLI with this command:

```
/opt/radiator/server/bin/radiator-client --auth-port 1812 --acct-port 1813
--secret mysecret --user <user@entra-domain.com> --password <password> --
timeout 120
```

Note: Replace `mysecret` with your configured RADIUS client secret, and use actual user credentials from your Entra ID tenant.

After running the test command, you should receive an MFA notification on your authenticator app:



MFA response options:

- **Accept the MFA** - Access will be granted and authentication succeeds
- **Reject the MFA** - Access will be denied and authentication fails
- **No response** - The request will timeout after the specified duration

The `--timeout 120` parameter allows sufficient time for users to respond to MFA prompts on their mobile devices.

UI

This configuration example has Radiator 10 GUI activated:

- Management UI is open to all IP addresses, protected by username and password or API key, configured in `/var/lib/radiator/management/` directory
- Use https://YOUR-IP-ADDRESS:8443/ to access
- Default username: *admin* and password: *ChangeThisPassword!*

Please note that Users and Clients cannot be added from the UI in this use case